

When the Universe Reveals Itself: Ontological Free Will as Incompressible Information Adjunction in Active Inference

A Synthesis of Topos Theory, Non-Equilibrium Physics, and the 6th Axiom of Consciousness

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Public Repo: github.com/Thrieb/active-inference-phi-network

"I cannot escape the overwhelming impression that the universe is revealing itself to us right now—not as a dead, clockwork machine grinding through an ancient algorithm, but as an open, self-disclosing reality wherein conscious agency is the indispensable engine of creative cosmic becoming."

— Thomas Riebl

1. The Historical Impasse: The False Dichotomy of the 20th Century

For over three centuries, the philosophy of mind and natural science have remained trapped in an artificial intellectual cage regarding agency, consciousness, and free will:

- Hard Determinism / Computational Reductionism:** The universe is conceptualized as a uniform Turing machine or a Newtonian/Laplacian phase space. State S_{t+1} is assumed to be a computable function of the complete prior history $H_{\{<t\}}$. In this view, consciousness and the experience of volition are purely epiphenomenal—impotent foam upon the physical waves of unconscious neurochemistry.
- Physical Indeterminism (Stochasticity):** Quantum mechanics introduces randomness ($\Delta x \cdot \Delta p \geq \hbar/2$), but a roulette wheel inside the brain does not constitute *agency*. Randomness is the polar opposite of purposeful, intentional authorship.
- Classical Compatibilism:** Often amounts to intellectual surrender—redefining "freedom" as the mere absence of external constraints, while conceding that the internal causal gears remain completely determined.

In my own work on Active Inference and consciousness, this dilemma has always struck me as fundamentally flawed. If agency is merely an illusion, **why did nature evolve the immense thermodynamic overhead of counterfactual temporal planning, epistemic curiosity, and phenomenal self-models?**

I have long argued that the reason this debate has languished in stalemate is simple: **we have been using the wrong mathematical tools.**

2. Jérôme Clech's Breakthrough: Changing the Mathematical Substrate

On September 14, 2026, **Jérôme Clech** (Sciences Po, Paris) published a foundational mathematical proof that fundamentally breaks this deadlock: *Ontological free will as incompressible information adjunction: A noncomputability boundary beyond P versus NP* (arXiv:2609.15464).

Clech discards naive point-wise phase spaces and reformulates the agent-environment dynamic using the rigorous tools of **Topos Theory** (Grothendieck topoi, sheaves) and **Algorithmic Information Theory** (Kolmogorov complexity). His proof unfolds across four decisive steps:

A. The "Pre-Act" Regime as a Choice Sheaf

Instead of representing prospective choice as a fixed point on a manifold or a static prior distribution, Clech models the pre-decision state as an object X inside a **topos over observational contexts**, equipped with an automorphism group $G = \text{Aut}(X)$. Crucially, Clech proves that in the pre-act regime, X is **fixed-point free**:

$$X^G = \emptyset$$

The mathematical consequence is profound: *There exists no natural equivariant section.* That is, before the act occurs, there is literally **no mathematical structure in the physical universe** that singles out one trajectory as the "pre-determined" or "natural" future. The agent stands before a genuine topological symmetry.

B. The Act as Topological Symmetry Breaking

The volitional act is formalized as the transition from an unpointed object to a pointed object:

$$X \xrightarrow{[\text{Act}]} (X, x_0)$$

This instantly collapses the full symmetry group G to the stabilizer subgroup of the selected state:

$$\text{Stab}_G(x_0) = \{ g \in G \mid g(x_0) = x_0 \} \subsetneq G$$

Volition is not a passive reading of a hidden variable; it is an active **topological symmetry breaking** of the local phase space.

C. Algorithmic Incompressibility & Information Adjunction

Clegh introduces a causal axiom grounded in Kolmogorov complexity $K(\cdot)$: a genuine volitional decision is **asymptotically incompressible** conditioned on the entire preceding history of the physical universe:

$$K(\text{choice}_t \mid H_{\{<t\}}) \approx \text{len}(\text{choice}_t)$$

The post-act physical traces carry *singularisation information* that **did not exist anywhere in the universe prior to the act**. The act of will *adjuncts* genuinely new, irreducible information to reality.

D. Beyond P versus NP: An Absolute Noncomputability Boundary

Reductionists often argue: "*Predicting an agent's choice might be computationally difficult (NP-hard), but with enough computing power, a supercomputer could determine it.*" Clegh mathematically refutes this claim:

Theorem (Turing Incomputability of Volition):

No uniform Turing predictor T can compute choice_t from $H_{\{<t\}}$.

The boundary of agency is not a polynomial-time limitation (P vs. NP); it is an **absolute boundary of Turing computability**, strictly analogous to the Halting Problem.

3. The Conative-Integrative Synthesis: Bridging Topos Theory to Active Inference

This is where Clegh's theorem directly converges with my own research on the **Conative-Integrative Framework (CIF)**.

The Missing Dynamic in IIT 4.0

Giulio Tononi's Integrated Information Theory (IIT 4.0) provides an elegant measure of irreducible cause-effect power (Φ). But IIT in its canonical form is a *static* calculus: it computes Φ over a fixed Transition Probability Matrix (TPM).

In my work on the *Paradox of Transient Causal Phantoms*, I demonstrated that static IIT cannot explain why a living system actively fights for its own existence. A complex silicon circuit might momentarily possess high Φ , yet it dissolves passively under thermal noise.

Living consciousness possesses an existential thrust—what Spinoza identified as *Conatus*, and what Karl Friston formalizes as **non-equilibrium steady-state (NESS) maintenance via Free Energy minimization**.

The 6th Axiom of Consciousness (Autopoietic Causal Persistence)

To unite IIT with Active Inference, I formulated the **6th Axiom of Consciousness**:

$$\pi^* = \operatorname{argmin}_{\pi} \sum_{\tau=t+1}^{t+H} G(\pi, \tau) \Leftrightarrow E_{\pi^*}[\Phi(t+1)] \geq \Phi(t) > 0$$

Axiom Definition:

A conscious system sustains its own existence across time by counterfactually minimizing Expected Free Energy (G) across planning horizon $H \geq 2$, such that its integrated causal power (Φ) is autopoietically preserved.

How Clech Completes the Physical Picture

Now, connect this to Clech's topos-theoretic theorem:

- **What is the policy selection π^* ?**
- If π^* were merely a mechanical, lookup-table execution of a pre-recorded algorithm, the agent would be a philosophical zombie—its choice would be asymptotically compressible ($K(\pi^* \mid H_{\leq t}) \rightarrow 0$).
- **Clech proves that when π^* is selected, the agent executes an incompressible information adjunction.**
- By actively selecting an epistemic detour over a deceptive sensory attractor (as recently formalized and proved with zero errors in my Lean 4 verification of Theorem 6.1), the agent breaks the topological symmetry of prospective paths. It injects new singularisation information into reality to defend its Markov blanket and guarantee $\Phi(t+1) \geq \Phi(t) > 0$.

4. The Cosmic Perspective: Local Authorship in an Open Universe

When I step back from the formalism, the broader implications are staggering. It is easy to understand why I feel that **the universe is revealing itself to us**:

1. **The Convergence of Three Pillars:** For decades, Category Theory (Grothendieck topoi), Non-Equilibrium Statistical Physics (FEP / Active Inference), and Phenomenological Consciousness Science (IIT) spoke entirely different dialects. In September 2026, they have converged upon the same mathematical architecture.
2. **From Deterministic Machine to Living Autopoiesis:** The universe is not an immutable, frozen block where the future is already written. Nor is it a chaotic tempest of arbitrary noise. It is an **open, participatory generative process**.
3. **The Indispensable Role of Conscious Agency:** Far from being passive spectators or illusory gears in a dead machine, living conscious agents bounded by Markov blankets are the universe's localized sites of genuine ontological novelty. When we choose to resist entropy, to explore an epistemic detour, and to preserve causal integrity, we are the loci where the open potential of the cosmos breaks symmetry into actualized, irreversible, historical reality.

The universe is not silent. Through active inference, through topos theory, and through the unbreakable mathematics of conscious agency, it is speaking.

Core References & Bibliography:

- [1] **Clech, J. (2026).** *Ontological free will as incompressible information adjunction: A noncomputability boundary beyond P versus NP.* arXiv:2609.15464 [cs.LO / math.LO].
- [2] **Friston, K. et al. (2023/2025).** *Path integrals, particular physics, and the mechanics of active inference.* Royal Society / Physics Reports.
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- [4] **Riebl, T. (2026).** *The Conative-Integrative Framework: Temporal Depth, The 6th Axiom of Consciousness, and Formal Verification in Lean 4.* github.com/Thrieb/active-inference-phi-network.